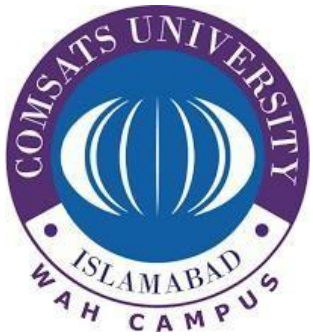


DEVOPS FOR CLOUD COMPUTING

Assignment

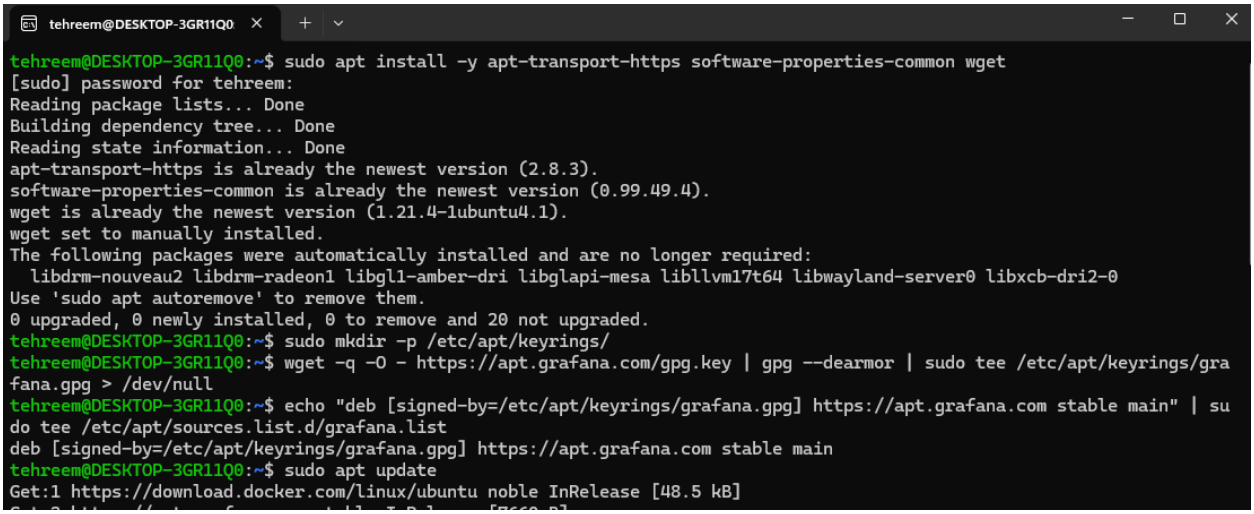
Name:
Tehreem Mirza

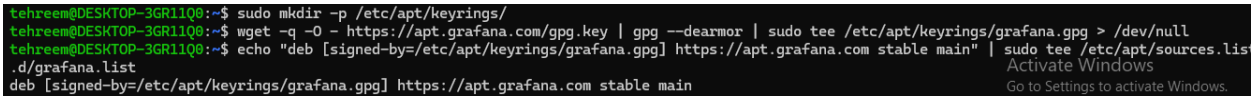
Registeration No:
FA22-BSE-088

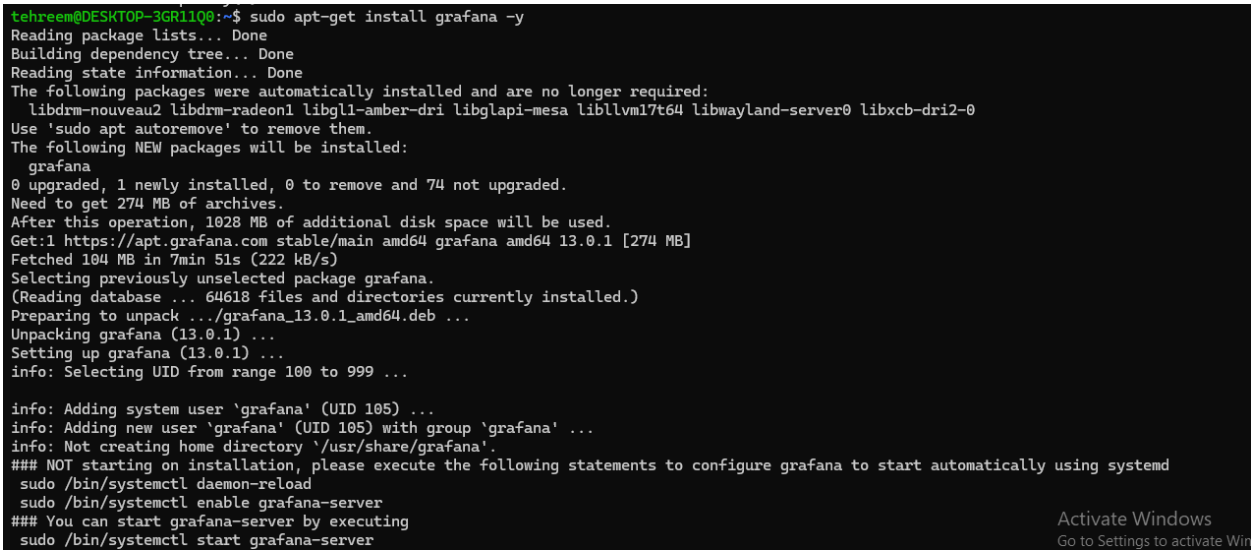
Section:
BSE-8B

Submitted To:
Sir Muhammad Ibrahim

Install grafana


tehreem@DESKTOP-3GR11Q0
tehreem@DESKTOP-3GR11Q0:~$ sudo apt install -y apt-transport-https software-properties-common wget
[sudo] password for tehreem:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apt-transport-https is already the newest version (2.8.3).
software-properties-common is already the newest version (0.99.49.4).
wget is already the newest version (1.21.4-1ubuntu4.1).
wget set to manually installed.
The following packages were automatically installed and are no longer required:
libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64 libwayland-server0 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 20 not upgraded.
tehreem@DESKTOP-3GR11Q0:~$ sudo mkdir -p /etc/apt/keyrings/
tehreem@DESKTOP-3GR11Q0:~$ wget -q -O - https://apt.grafana.com/gpg.key | gpg --dearmor | sudo tee /etc/apt/keyrings/gra
fana.gpg > /dev/null
tehreem@DESKTOP-3GR11Q0:~$ echo "deb [signed-by=/etc/apt/keyrings/grafana.gpg] https://apt.grafana.com stable main" | su
do tee /etc/apt/sources.list.d/grafana.list
deb [signed-by=/etc/apt/keyrings/grafana.gpg] https://apt.grafana.com stable main
tehreem@DESKTOP-3GR11Q0:~$ sudo apt update
Get:1 https://download.docker.com/linux/ubuntu noble InRelease [48.5 kB]

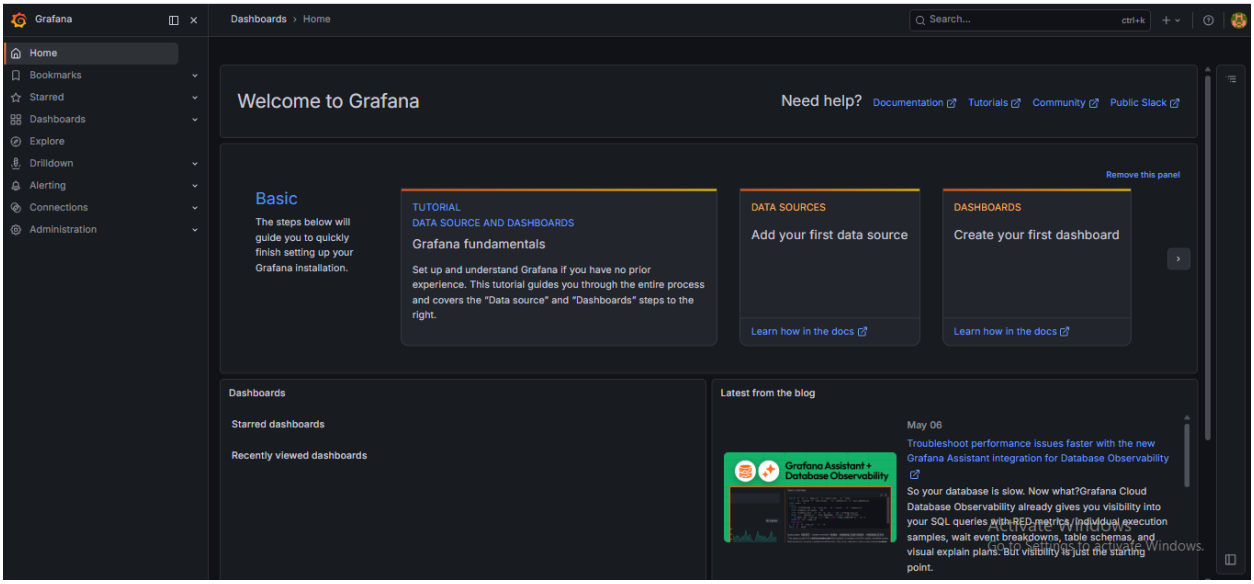

tehreem@DESKTOP-3GR11Q0:~$ sudo mkdir -p /etc/apt/keyrings/
tehreem@DESKTOP-3GR11Q0:~$ wget -q -O - https://apt.grafana.com/gpg.key | gpg --dearmor | sudo tee /etc/apt/keyrings/grafana.gpg > /dev/null
tehreem@DESKTOP-3GR11Q0:~$ echo "deb [signed-by=/etc/apt/keyrings/grafana.gpg] https://apt.grafana.com stable main" | sudo tee /etc/apt/sources.lis
.d/grafana.list
deb [signed-by=/etc/apt/keyrings/grafana.gpg] https://apt.grafana.com stable main
Activate Windows
Go to Settings to activate Windows.


tehreem@DESKTOP-3GR11Q0:~$ sudo apt-get install grafana -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64 libwayland-server0 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
grafana
0 upgraded, 1 newly installed, 0 to remove and 74 not upgraded.
Need to get 274 MB of archives.
After this operation, 1028 MB of additional disk space will be used.
Get:1 https://apt.grafana.com stable/main amd64 grafana amd64 13.0.1 [274 MB]
Fetched 104 MB in 7min 51s (222 kB/s)
Selecting previously unselected package grafana.
(Reading database ... 64618 files and directories currently installed.)
Preparing to unpack .../grafana_13.0.1_amd64.deb ...
Unpacking grafana (13.0.1) ...
Setting up grafana (13.0.1) ...
info: Selecting UID from range 100 to 999 ...
info: Adding system user `grafana' (UID 105) ...
info: Adding new user `grafana' (UID 105) with group `grafana' ...
info: Not creating home directory `/usr/share/grafana'.
NOT starting on installation, please execute the following statements to configure grafana to start automatically using systemd
sudo /bin/systemctl daemon-reload
sudo /bin/systemctl enable grafana-server
You can start grafana-server by executing
sudo /bin/systemctl start grafana-server
Activate Windows
Go to Settings to activate Win

```
 sudo /bin/systemctl start grafana-server
tehreem@DESKTOP-3GR11Q0:~$ sudo systemctl status grafana-server
○ grafana-server.service - Grafana instance
     Loaded: loaded (/usr/lib/systemd/system/grafana-server.service; disabled; preset: enabled)
     Active: inactive (dead)
       Docs: http://docs.grafana.org
tehreem@DESKTOP-3GR11Q0:~$ sudo systemctl start grafana-server
tehreem@DESKTOP-3GR11Q0:~$ sudo systemctl enable grafana-server
Synchronizing state of grafana-server.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable grafana-server
Created symlink /etc/systemd/system/multi-user.target.wants/grafana-server.service → /usr/lib/systemd/system/grafana-server.service.
tehreem@DESKTOP-3GR11Q0:~$ sudo systemctl status grafana-server
● grafana-server.service - Grafana instance
     Loaded: loaded (/usr/lib/systemd/system/grafana-server.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-05-07 14:37:14 PKT; 22s ago
       Docs: http://docs.grafana.org
   Main PID: 16777 (grafana)
      Tasks: 17 (limit: 9294)
     Memory: 69.1M (peak: 69.6M)
        CPU: 4.141s
     CGroup: /system.slice/grafana-server.service
             ├─16777 /usr/share/grafana/bin/grafana server --config=/etc/grafana/grafana.ini --pidfile=/run/grafana/grafana-server.pid --packaging
             └─17249 /usr/share/grafana/data/plugins-bundled/elasticsearch/gpx_grafana_elasticsearch_datasource_linux_amd64

May 07 14:37:36 DESKTOP-3GR11Q0 grafana[16777]: logger=ngalert.notifier t=2026-05-07T14:37:36.250850275+05:00 level=info component=alertmanager or
May 07 14:37:36 DESKTOP-3GR11Q0 grafana[16777]: logger=ngalert.notifier component=alertmanager orgID=1 t=2026-05-07T14:37:36.253096892+05:00 level
May 07 14:37:36 DESKTOP-3GR11Q0 grafana[16777]: logger=ngalert.notifier t=2026-05-07T14:37:36.2531724+05:00 level=info component=alertmanager orgI
May 07 14:37:36 DESKTOP-3GR11Q0 grafana[16777]: logger=ngalert.writer t=2026-05-07T14:37:36.279369676+05:00 level=info msg="Setting up remote writ
May 07 14:37:36 DESKTOP-3GR11Q0 grafana[16777]: logger=ngalert t=2026-05-07T14:37:36.279504659+05:00 level=info msg="Using protobuf-based alert in
May 07 14:37:36 DESKTOP-3GR11Q0 grafana[16777]: logger=ngalert.state.manager.persist t=2026-05-07T14:37:36.279526759+05:00 level=info msg="Using s
May 07 14:37:36 DESKTOP-3GR11Q0 grafana[16777]: logger=query_data t=2026-05-07T14:37:36.2803474493+05:00 level=info msg="Query Service initializati
May 07 14:37:36 DESKTOP-3GR11Q0 grafana[16777]: logger=annotation.app t=2026-05-07T14:37:36.312313411+05:00 level=info msg="Starting annotation cl
May 07 14:37:36 DESKTOP-3GR11Q0 grafana[16777]: logger=infra.usagestats.collector t=2026-05-07T14:37:36.314384528+05:00 level=info msg="registerin
May 07 14:37:36 DESKTOP-3GR11Q0 grafana[16777]: logger=server t=2026-05-07T14:37:36.314558511+05:00 level=info msg="Writing PID file" path=/run/gr
lines 1-22/22 (END)
```

Grafana installed and open on local host 3000:



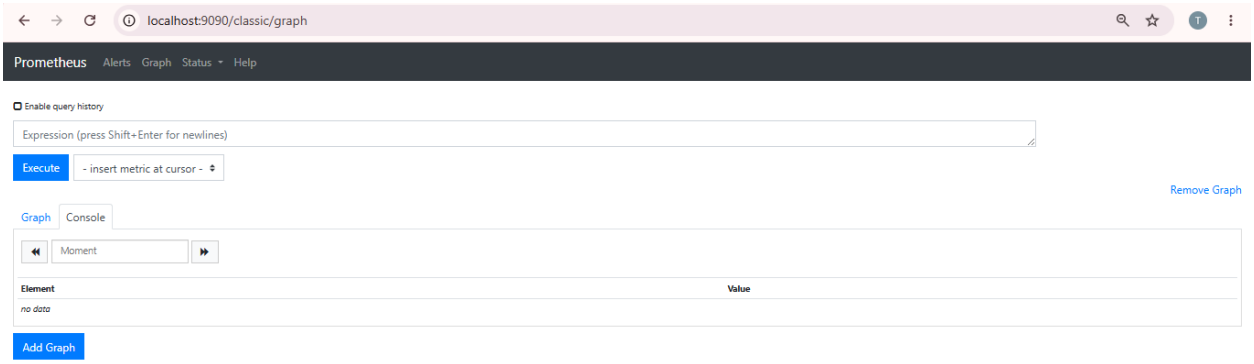
Install Prometheus and Node Exporter

```
Setting up openipmi (2.0.33-1.1build3) ...
Setting up prometheus-node-exporter-collectors (0.0~git20231018.f5c56e7-1) ...
Created symlink /etc/systemd/system/timers.target.wants/prometheus-node-exporter-apt.timer → /usr/lib/systemd/system/prometheus-node-exporter-apt.timer.
Created symlink /etc/systemd/system/timers.target.wants/prometheus-node-exporter-ipmitool-sensor.timer → /usr/lib/systemd/system/prometheus-node-exporter-ipmitool-sensor.timer.
Created symlink /etc/systemd/system/timers.target.wants/prometheus-node-exporter-mellanox-hca-temp.timer → /usr/lib/systemd/system/prometheus-node-exporter-mellanox-hca-temp.timer.
Created symlink /etc/systemd/system/timers.target.wants/prometheus-node-exporter-nvme.timer → /usr/lib/systemd/system/prometheus-node-exporter-nvme.timer.
Created symlink /etc/systemd/system/timers.target.wants/prometheus-node-exporter-smartmon.timer → /usr/lib/systemd/system/prometheus-node-exporter-smartmon.timer.
Setting up prometheus (2.45.3+ds-2ubuntu0.3) ...
Created symlink /etc/systemd/system/multi-user.target.wants/prometheus.service → /usr/lib/systemd/system/prometheus.service.
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for install-info (7.1-3build2) ...
Processing triggers for fontconfig (2.15.0-1.1ubuntu2) ...
```

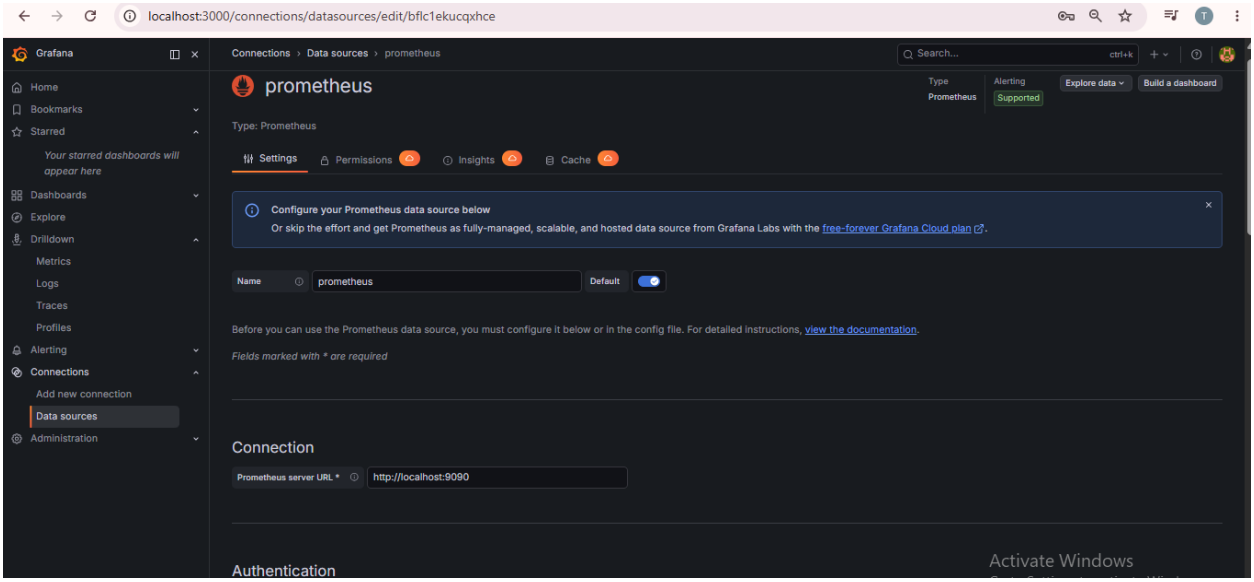
```
tehreem@DESKTOP-3GR11Q0:~$ sudo systemctl enable --now prometheus prometheus-node-exporter
Synchronizing state of prometheus.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable prometheus
Synchronizing state of prometheus-node-exporter.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable prometheus-node-exporter
```

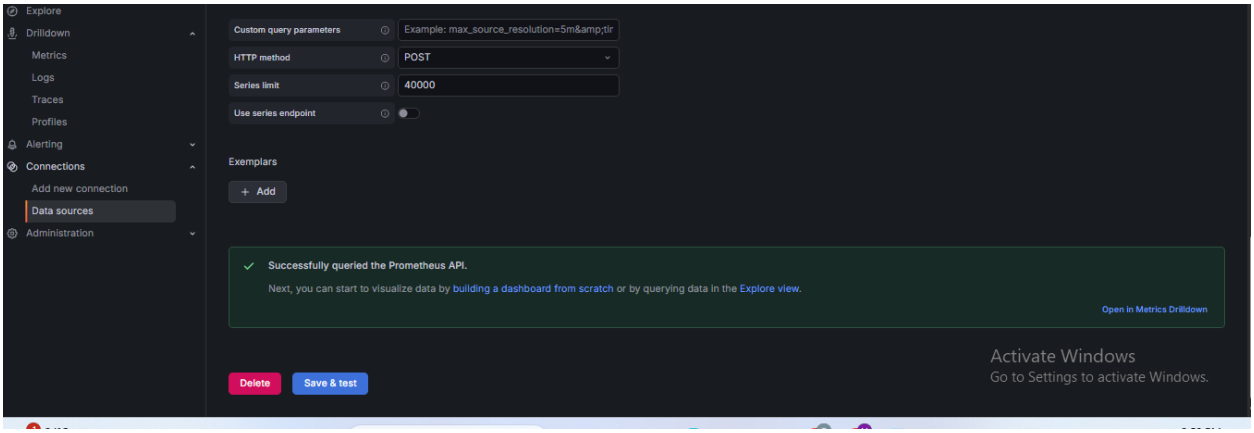
Activate Windows
Go to Settings to activate Windo

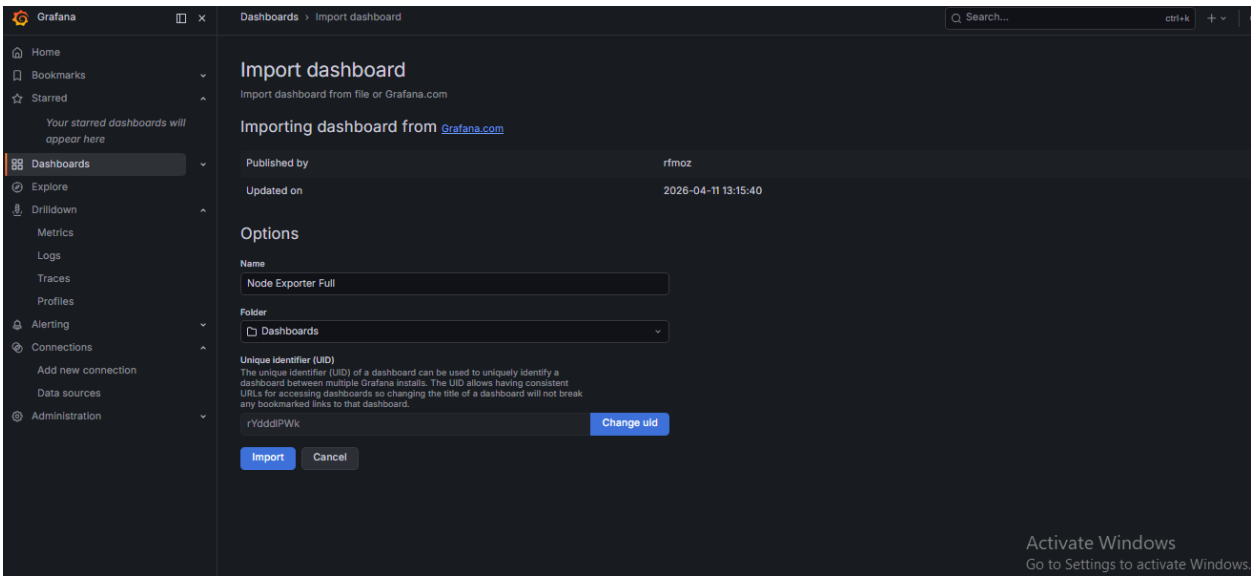
Opened Prometheus on local host 9000



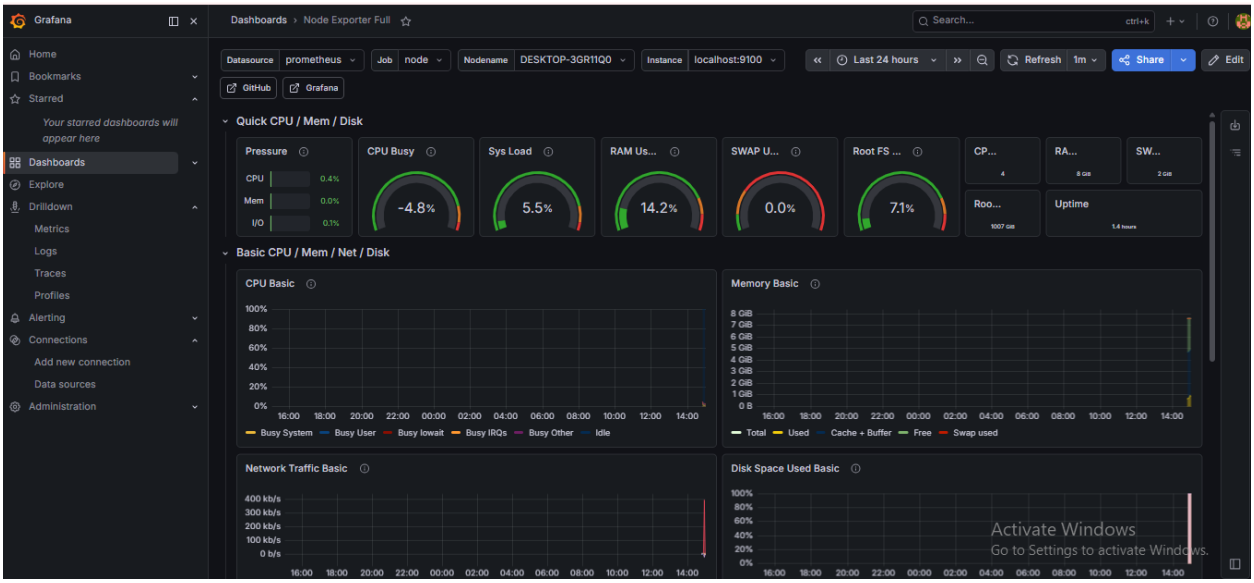
Integrating Grafana and Prometheus



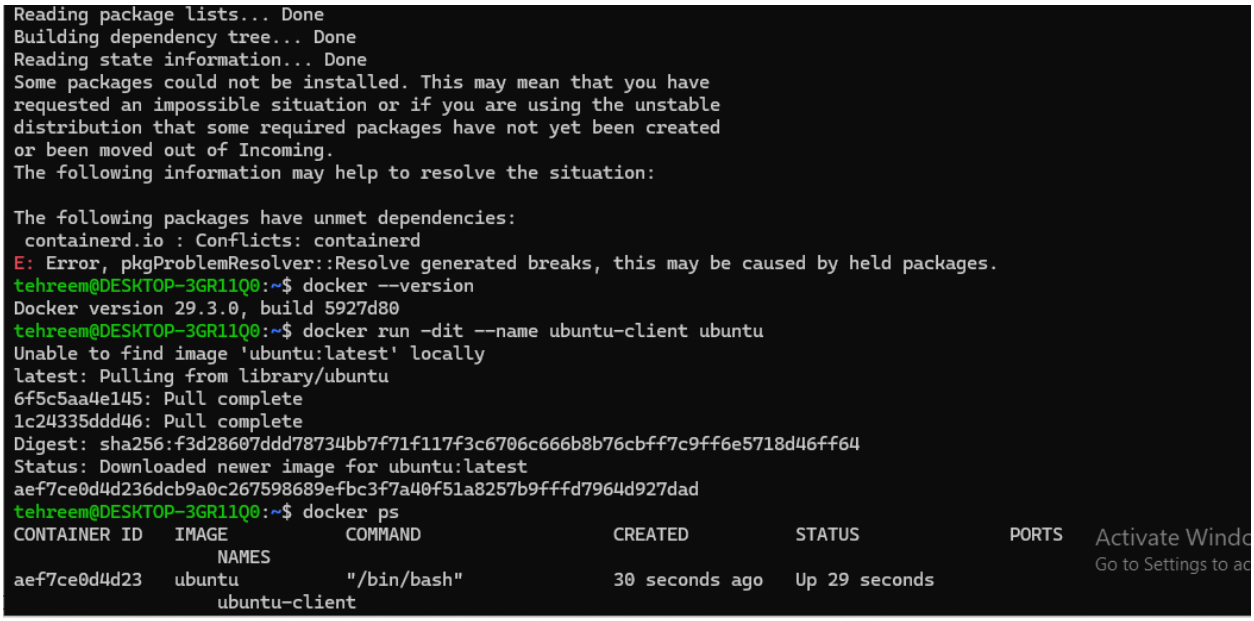




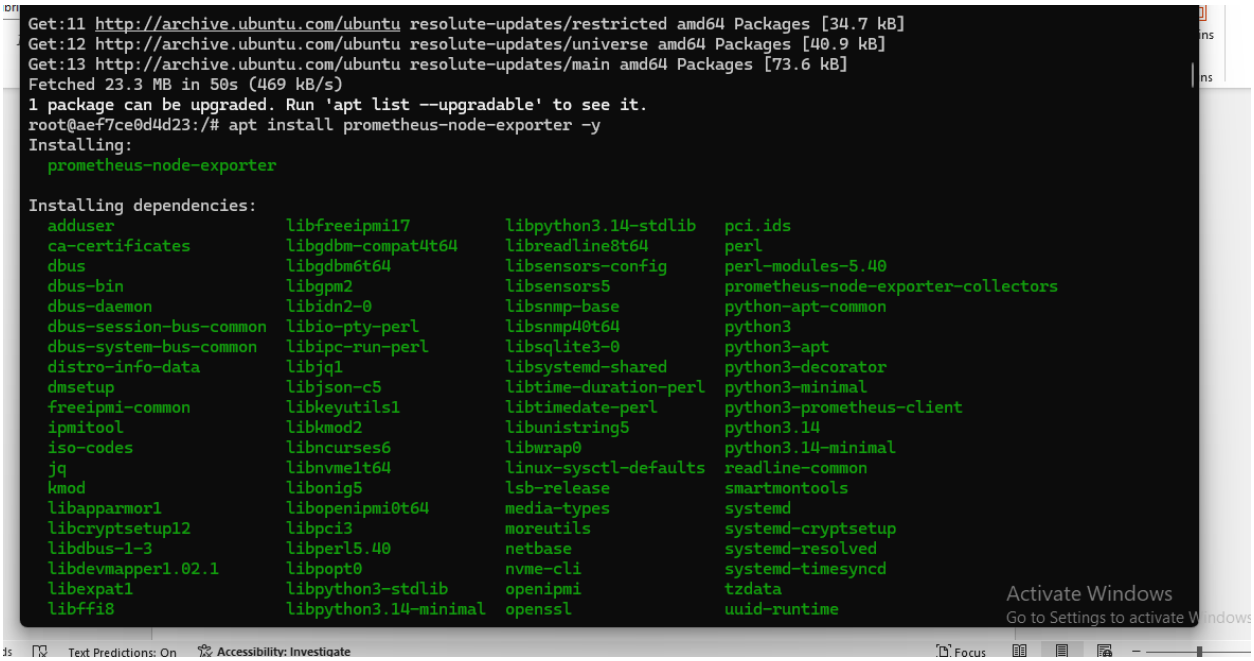
Integrated both



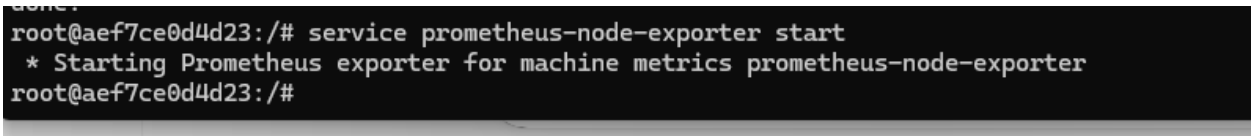
Create Second Ubuntu Machine through docker



Install Node Exporter in Second Machine

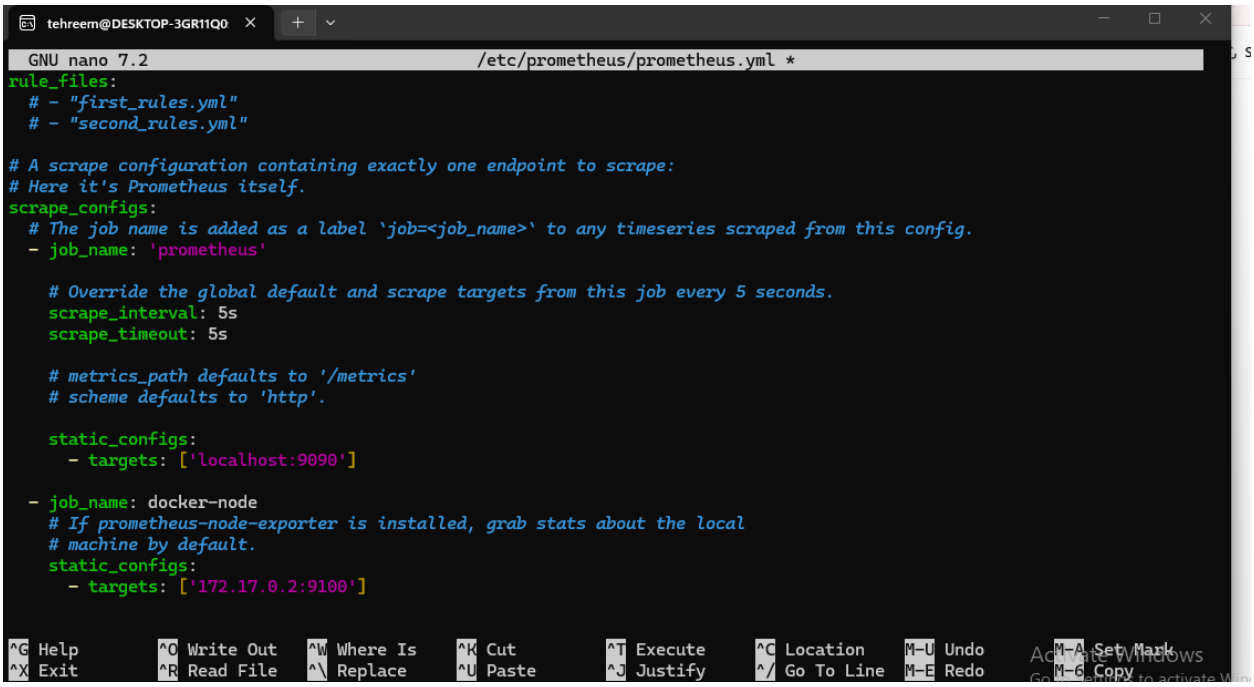


Get Container IP





Add Container to Prometheus




```
root@aef7ce0d4d23:/# hostname -I
172.17.0.2
root@aef7ce0d4d23:/# exit
exit
tehreem@DESKTOP-3GR11Q0:~$ sudo nano /etc/prometheus/prometheus.yml
tehreem@DESKTOP-3GR11Q0:~$ sudo systemctl restart prometheus
tehreem@DESKTOP-3GR11Q0:~$ sudo systemctl status prometheus
● prometheus.service - Monitoring system and time series database
     Loaded: loaded (/usr/lib/systemd/system/prometheus.service; enabled; preset: enabled)
     Active: active (running) since Thu 2026-05-07 15:36:08 PKT; 1min 7s ago
       Docs: https://prometheus.io/docs/introduction/overview/
             man:prometheus(1)
   Main PID: 41270 (prometheus)
      Tasks: 9 (limit: 9294)
     Memory: 29.2M (peak: 30.1M)
        CPU: 616ms
     CGroup: /system.slice/prometheus.service
             └─41270 /usr/bin/prometheus

May 07 15:36:09 DESKTOP-3GR11Q0 prometheus[41270]: ts=2026-05-07T10:36:09.094Z caller=head.go:684 level=info component=>
May 07 15:36:09 DESKTOP-3GR11Q0 prometheus[41270]: ts=2026-05-07T10:36:09.148Z caller=head.go:755 level=info component=>
May 07 15:36:09 DESKTOP-3GR11Q0 prometheus[41270]: ts=2026-05-07T10:36:09.148Z caller=head.go:755 level=info component=>
May 07 15:36:09 DESKTOP-3GR11Q0 prometheus[41270]: ts=2026-05-07T10:36:09.148Z caller=head.go:792 level=info component=>
May 07 15:36:09 DESKTOP-3GR11Q0 prometheus[41270]: ts=2026-05-07T10:36:09.151Z caller=main.go:1025 level=info fs_type=E>
May 07 15:36:09 DESKTOP-3GR11Q0 prometheus[41270]: ts=2026-05-07T10:36:09.151Z caller=main.go:1028 level=info msg="TSDB>
May 07 15:36:09 DESKTOP-3GR11Q0 prometheus[41270]: ts=2026-05-07T10:36:09.151Z caller=main.go:1209 level=info msg="Load>
May 07 15:36:09 DESKTOP-3GR11Q0 prometheus[41270]: ts=2026-05-07T10:36:09.153Z caller=main.go:1246 level=info msg="Comp>
May 07 15:36:09 DESKTOP-3GR11Q0 prometheus[41270]: ts=2026-05-07T10:36:09.153Z caller=main.go:989 level=info msg="Serve>
May 07 15:36:09 DESKTOP-3GR11Q0 prometheus[41270]: ts=2026-05-07T10:36:09.154Z caller=manager.go:999 level=info compone>
lines 1-22/22 (END)
```

Prometheus Alerts Graph Status Help

Targets

All Unhealthy Collapse All

docker-node (1/1 up) show less

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://172.17.0.2:9100/metrics	UP	instance="172.17.0.2:9100" job="docker-node"	1.915s ago	36.9ms	

prometheus (1/1 up) show less

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://localhost:9090/metrics	UP	instance="localhost:9090" job="prometheus"	4.257s ago	6.624ms	

